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-----------------------------------------------------------------------------
NAME:
LASTNAME:
STUDENT#:
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INSTRUCTIONS:
For this exam, only the use of a basic text editor (e.g. gedit, kwrite) is allowed.
ALl other software applications must be closes.
You may activate C++ highlighting option in case that is helpful.

Answer the questions briefly: maximum two to three paragraphs of a few sentences.
Use the correct terminology to formulate your answer as clearly as possible.
When a code sample/fragment is required, make it as concise as possible:
- The '#includes' of header files can be ignored.
- Keep function, class and variable names as short as possible.
- You do, however, respect the identification rules.


When you are done with the exam:
- Save your file with the name "LASTNAME-NAME_studentNumber_advProg-Examen-22-23-
1ezit.txt" with your corresponding details.
- Please contact the supervisor for further instructions.



%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
1- Basic Facilities (3)
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%
Question: From the point of view of semantics, describe two differences between a
"Static" and "Dynamic" property.
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%----------------------------------------------------------------------------
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Question: Describe the sequence of conversions considered when doing Overloading
Resolution.
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%----------------------------------------------------------------------------
%
Question: Describe the difference between "Inline Functions and "Consexpr
Functions".
%


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
2- Classes (1)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
Question: Describe the difference between physical and logical constness. Indicate
a scenario where each of these types of constness are desirable.
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%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
3- Construction & Cleanup (2)
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Question: Describe how the construction and destruction operations take place for
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objects of derived classes.
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%
Question: Describe what we mean by Resource Acquisition is Initialization (RAII).
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%%
4- Operator Overloading (1)
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%
Describe the restrictions that are in place when overloading operators.
%

%%
5- Copy & Move (1)
%%
%
Question: Describe the requirements that the implementation of a "copy constructor" and a "copy assignment" should possess in order to be considered a valid copy operation. Indicate a scenario where these requirements are violated.
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%%
6- Class Derivation and Inheritance (1)
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%
Question: Describe what is "virtual table". Indicate what is its function and how it is used in the context of class derivation and inheritance.
%

%%
7- Object-Oriented Programming (1)
%%
%
Question: Describe what do we mean by "Type Conformity". Further indicate how this is related with "classes" and "types".
%

%%
8- Design Patterns (1)
%%
%
Question: Describe the "Observer" design pattern. Indicate why this pattern is considered a "Behavioral" pattern
%

%%
9- Generic Programming & Templates (1)
%%
%
Question: What is the main difference between "explicit specification" and "deduction via call arguments" when conducting "template argument deduction"?

Provide a code fragment displaying these two cases.
%